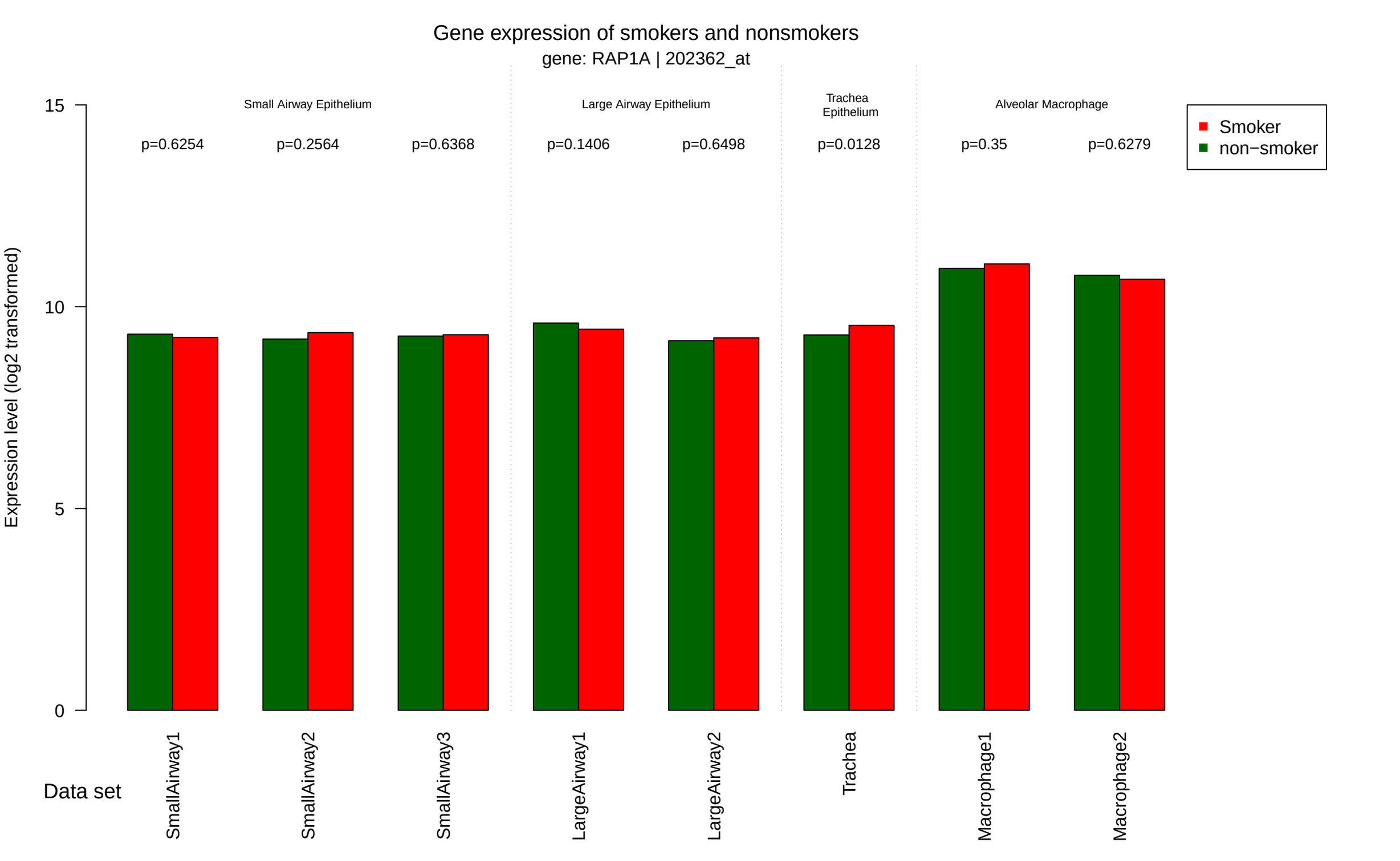
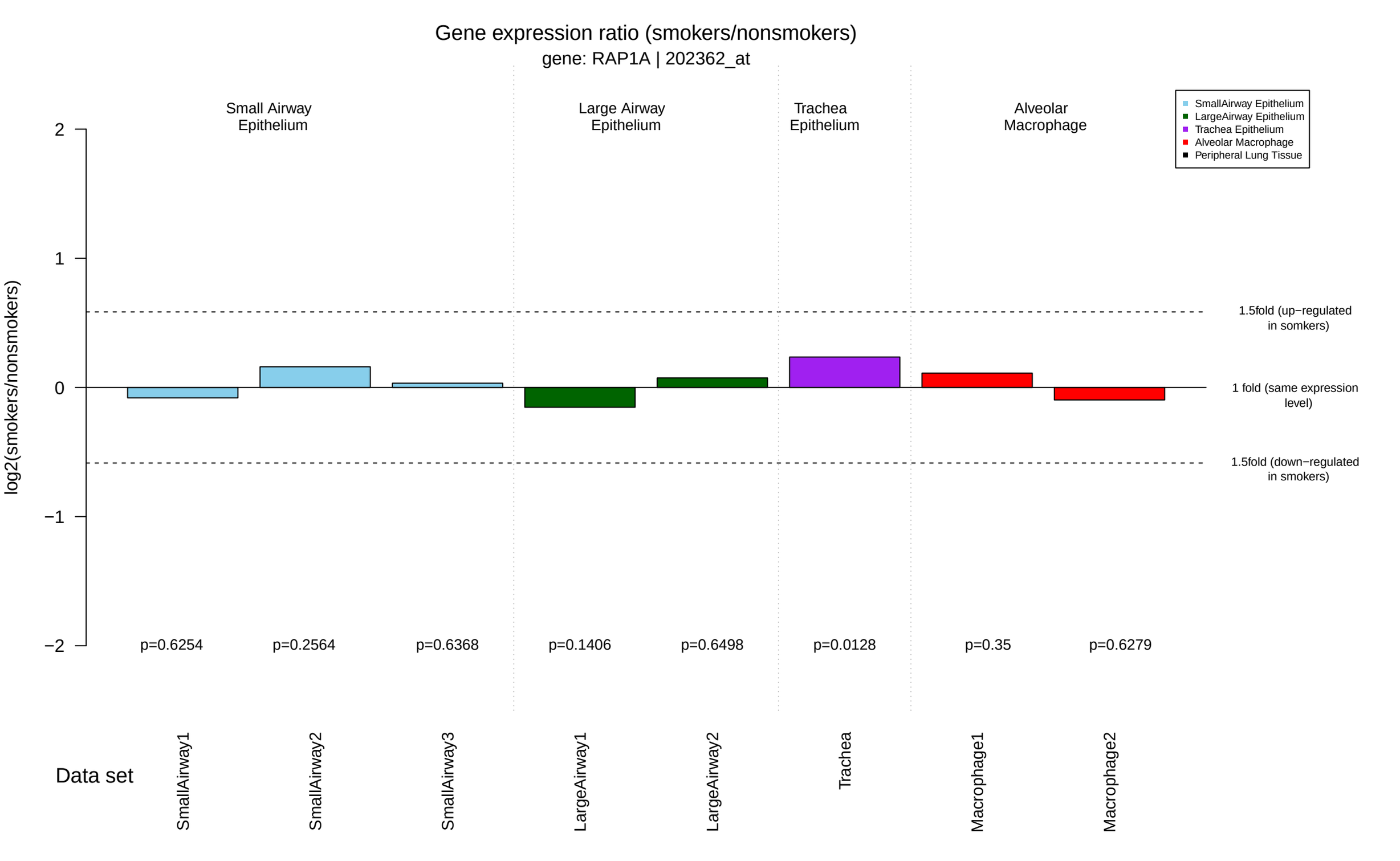
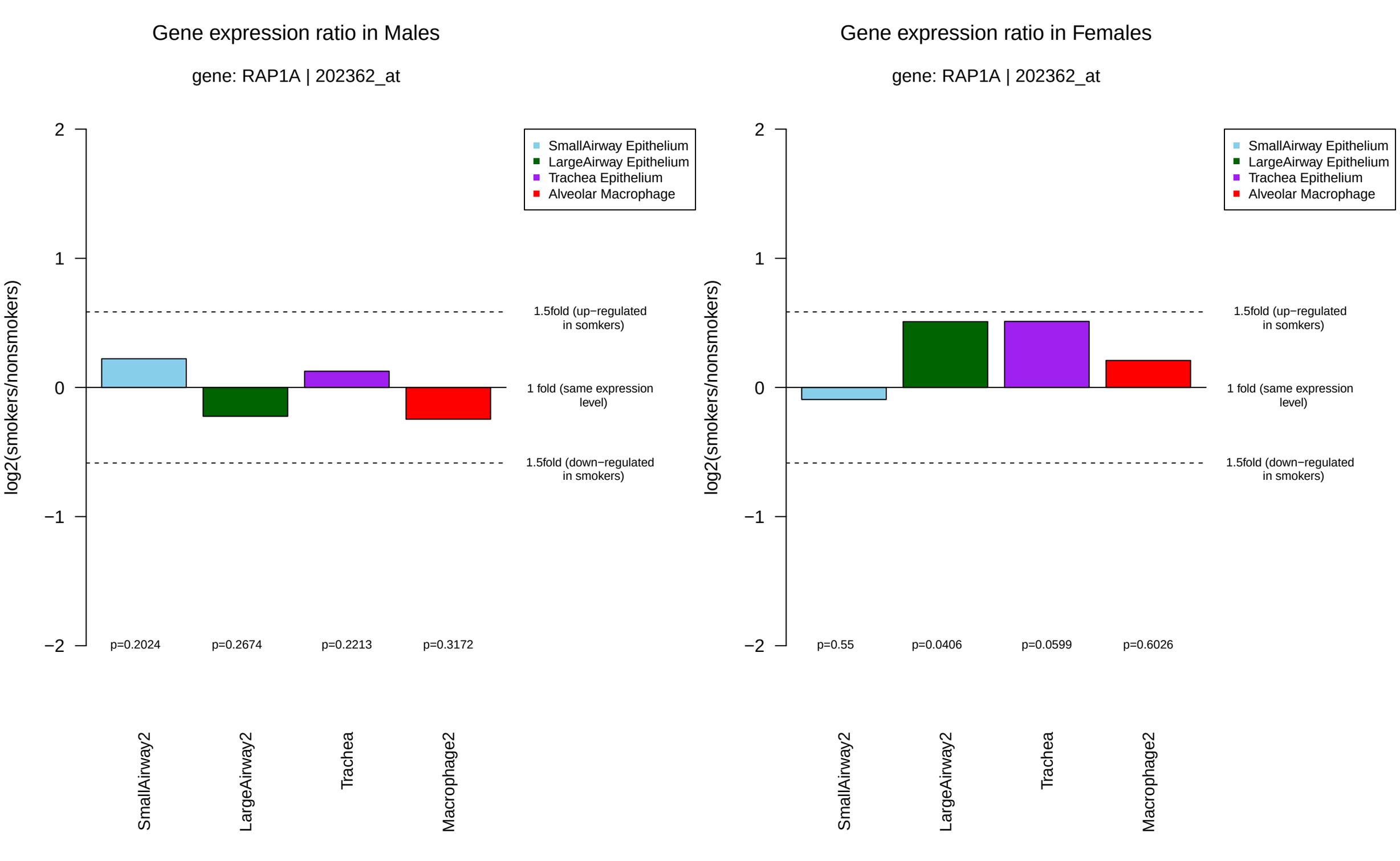
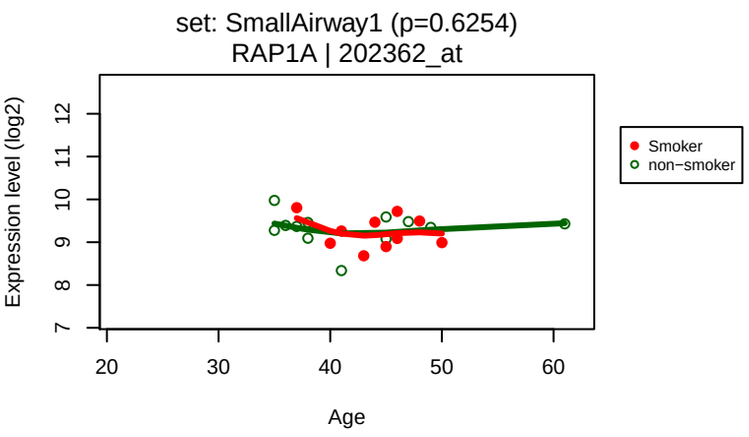
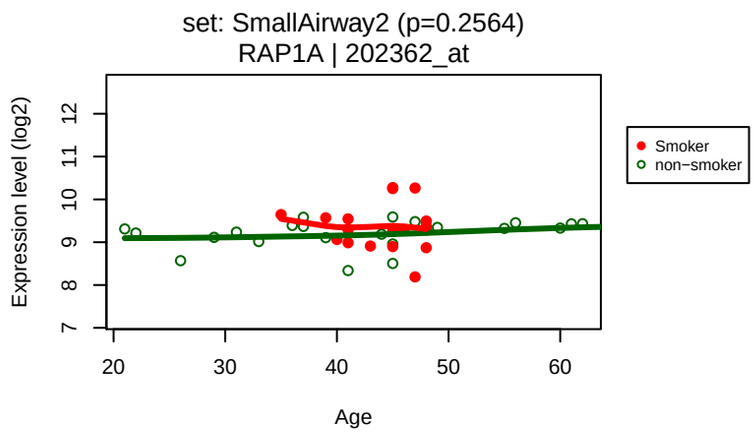
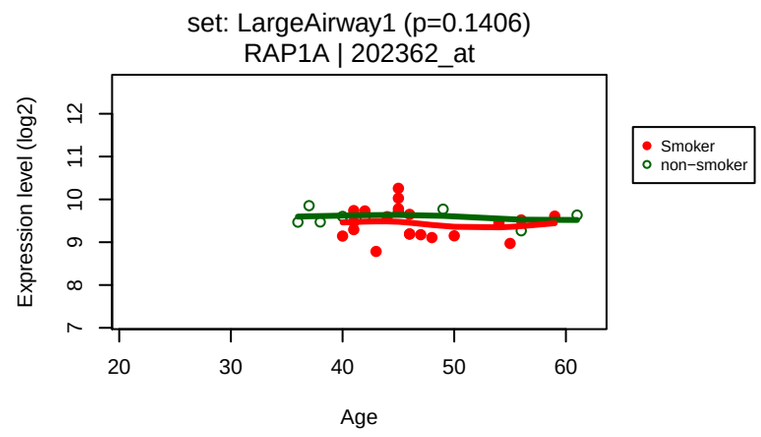

Gene expression of smokers and nonsmokers
gene: RAP1A | 202362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6254
p=0.2564
p=0.6368
p=0.1406
p=0.6498
p=0.0128
p=0.35
p=0.6279
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

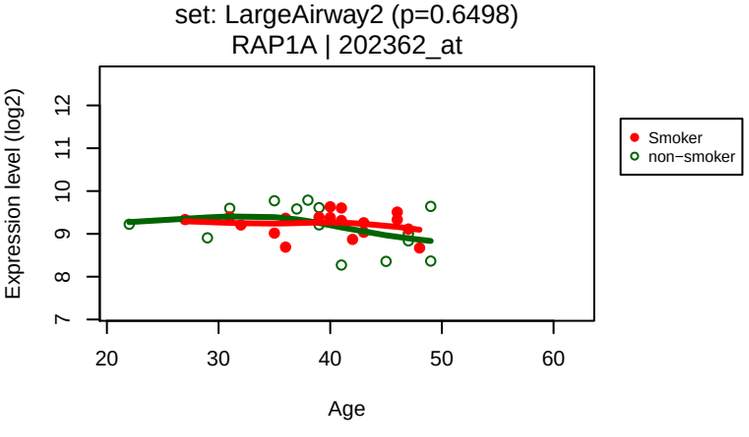

Gene expression ratio (smokers/nonsmokers)
gene: RAP1A | 202362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6254
p=0.2564
p=0.6368
p=0.1406
p=0.6498
p=0.0128
p=0.35
p=0.6279
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

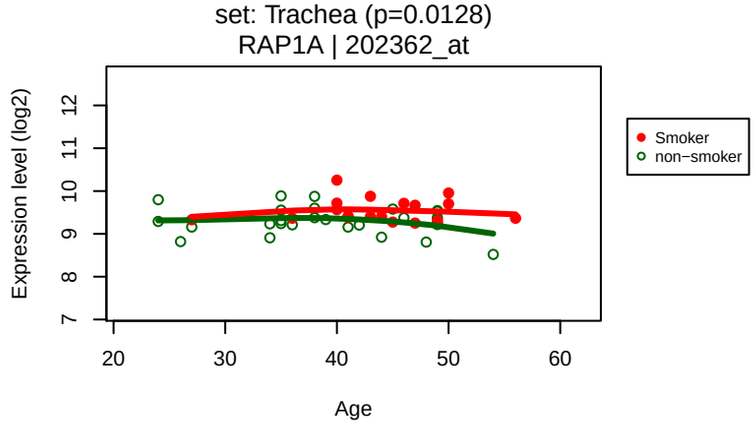

Gene expression ratio in Males
gene: RAP1A | 202362_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2024
p=0.2674
p=0.2213
p=0.3172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1A | 202362_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.55
p=0.0406
p=0.0599
p=0.6026
SmallAirway2
LargeAirway2
Trachea
Macrophage2

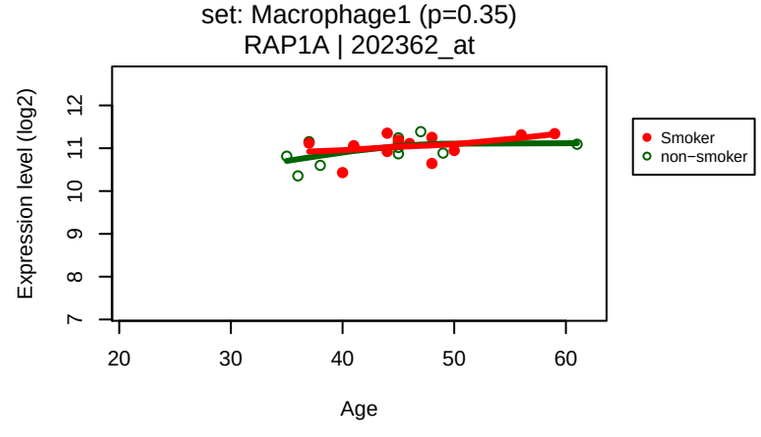

set: SmallAirway1 (p=0.6254)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker

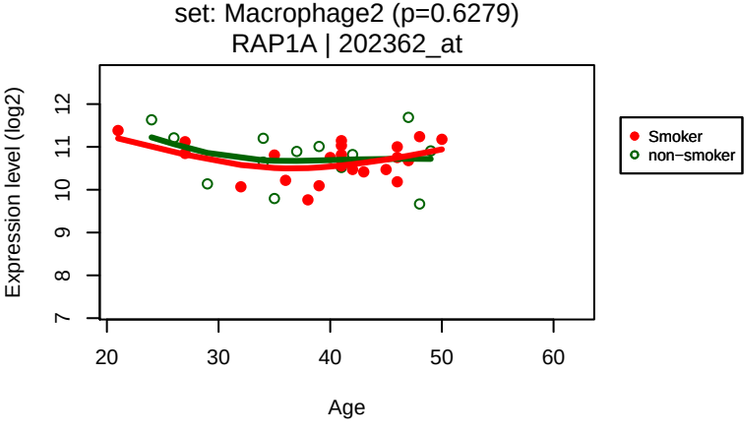

set: SmallAirway2 (p=0.2564)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker

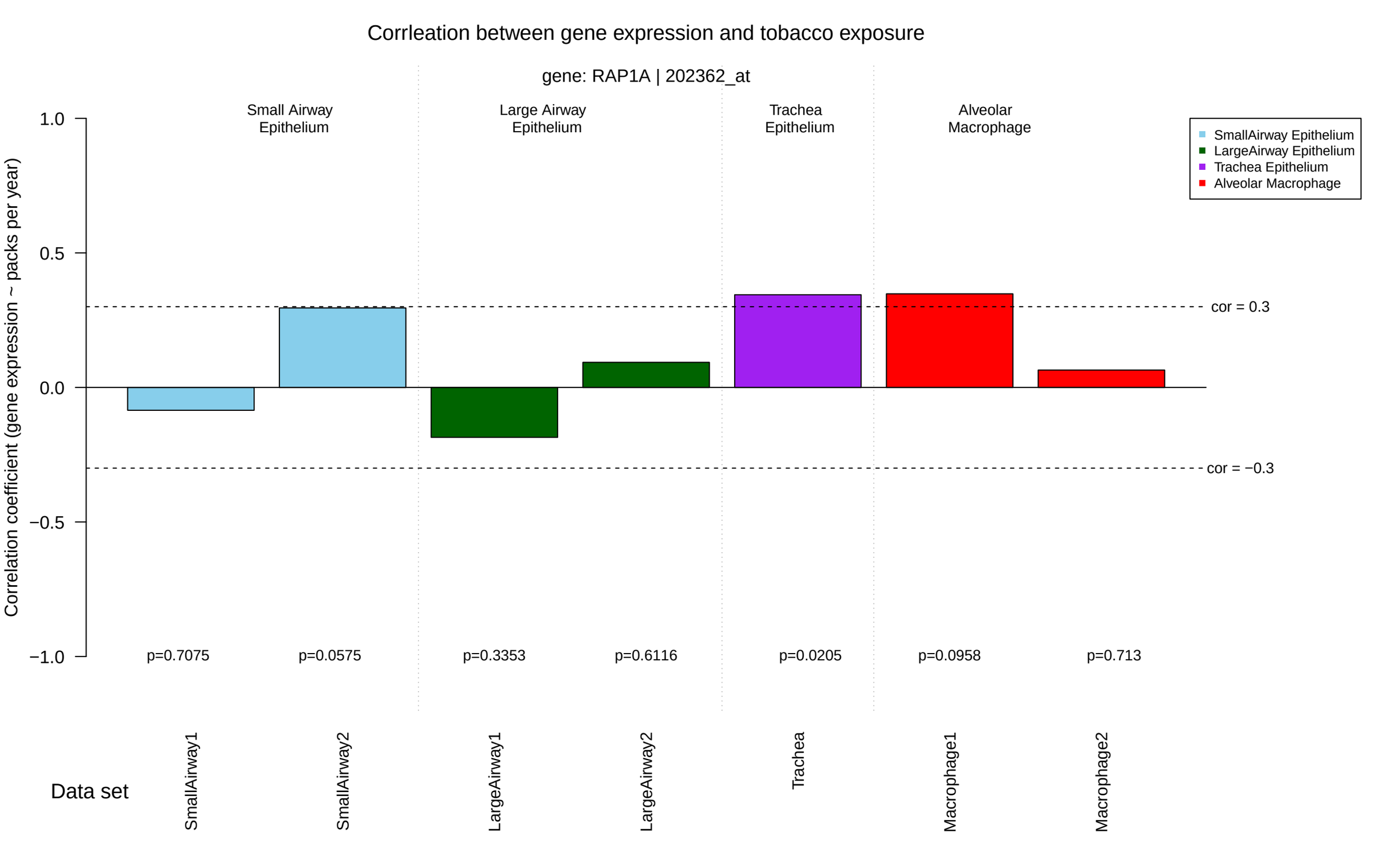

set: LargeAirway1 (p=0.1406)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6498)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0128)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.35)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6279)
RAP1A | 202362_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1A | 202362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7075
p=0.0575
p=0.3353
p=0.6116
p=0.0205
p=0.0958
p=0.713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2